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Phase I

Section A – Test Details					
Initiator:	America Alanis (Written by Qiyue Chen)	**Test Date:**	01-Sep-2026	**Date Initiated:**	08-Sep-2026
Date of Incident:	08-Sep-2026	**Test Name:**	Celsis Sterility	**Sample ID:**	ETX-260828-0527
Name of Analyst who Performed the Test:	Prepping Analyst: Andrew Carrillo (AC) Processing Analyst:	**Sample / Active Name**	MOTs-C 10 MG/ML (5 ML) Injection	**Lot #:**	LG342010349
		Dosage Form	Injection		
Description of Incident:	On 08 Sep 2026, sample ETX-260828-0527 was found to be positive for growth after Celsis		☑ Client Care notified of OOS ☑ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS		
[redacted]	SOP / Test Method # MICRO-SOP-44	Effective Date: 03-Jan-2025	SOP / Test Method Rev: 01	Limits/Specification: Pass/Fail	
Manager notified	Manager/designee name: Kathan Parikh	Date: 08-Sep-2026	Quality Assurance Department notified:	Yes ☑ No ☐	

Section B – Investigation and Root Cause Analysis			
Personnel Discrepancy		**Comments**	
Analyst interviewed?	☑Yes ☐ No ☐ N/A	Yes, analysts Andrew Carrillo,	
Correct samples analyzed?	☑Yes ☐ No ☐ N/A	Yes, Sample ID	
Correct SOP / Test Method used/followed?	☑Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-44	
Correct sample collection/technique?	☑Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-44	
Are the Test Result Records Attached & Calculations Verified and Correct?	☑Yes ☐ No ☐ N/A	Yes, Information is available in	
Is the analyst qualified to perform the test?	☑Yes ☐ No ☐ N/A	Yes, all analysts are trained and	
Is formula worksheet provided and verified?	☐Yes ☐ No ☑ N/A	Not Applicable	
Does the processing method/method set conform to the required specification?	☐Yes ☐ No ☑ N/A	Not Applicable	
Was the sample stored appropriately?	☑Yes ☐ No ☐ N/A	Yes, the sample was stored	
Material Discrepancy		**Lot Number**	**Expiry Date**
Is the integrity of the sample(s) questionable?	☐Yes ☑ No ☐N/A	[redacted]	
Was the sample transported appropriately?	☑Yes ☐ No ☐N/A		
Are the other test results from this run under investigation?	☐Yes ☑ No ☐N/A		

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Appropriate/correct glassware used?	☐Yes ☐ No ☑N/A		
Are reagents stored in proper condition and not expired?	☑Yes ☐ No ☐N/A	Materials and Reagents TSB: 06242026-1 FTM: 06162026-2 PBS: 35725002	Materials and Reagents TSB: 22 Sep 2026 FTM: 14 Sep 2026 PBS: 31 Dec 2027
Standard Used	Celsis ATP Positive Control	022601-1483	31 Jan 2027
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐Yes ☐ No ☑N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐Yes ☐ No ☑N/A		
Equipment Discrepancy		**Equipment ID**	**Calibration Due Date**
Was the instrument set up correctly?	☑Yes ☐ No ☐ N/A	E002222	Sep 2026
Is the clean room facility certified?	☑Yes ☐ No ☐ N/A	E001736 (CR	Dec 2026, Jun
System Performance Check (if applicable)?	☑Yes ☐ No ☐ N/A	E002222	Sep 2026
System Suitability RSD < 2% (PDA)?	☐Yes ☐ No ☑ N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐Yes ☐ No ☑ N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☑Yes ☐ No ☐ N/A	See Phase I	See Phase I
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☑Yes ☐ No ☐ N/A	See Phase I	See Phase I
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☑Yes ☐ No ☐ N/A	Incubator E001356 (Sensor E001450) Incubator E001357 (Sensor E001449)	Jan 2027 / Feb 2027 Jan 2027 / Feb 2027 Aug 2027 / Feb 2027 Aug 2027 / Feb

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[redacted]

Are chromatogram/spectra/raw data correct?	☐ Yes ☐ No ☑ N/A	Not Applicable	Not Applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)

- ☑ N/A
- ☐ Re-injection of sample vial
- ☐ Re-injection of original standard vial
- ☐ Re-vialing of original working standard solution
- ☐ Re-vialing of original working sample solution
- ☐ Re-dilution of a working sample solution from the original stock
- ☐ Re-dilution of a working standard solution from the original stock standard
- ☐ Other: N/A QYC 08Sep2026

Section D – Phase I Summary

Phase I Summary:

All analysts involved in the preparation, processing, aliquoting, and reading of the samples, Andrew Carrillo, Gabrielle Surber, America Alanis, and Cuong Du, were interviewed comprehensively. Their responses are documented throughout this report. This investigation was performed as per MICRO-SOP-53 Sterility Test Out-of-Specification (OOS) Investigation Procedure.

Upon arrival, the sample was stored refrigerated in accordance with the client instructions. Both sample prepping analyst Andrew Carrillo and processing analyst Gabrielle Surber inspected the sample during preparation and processing. No leaks or visible turbidity were observed, and no evidence of compromised container integrity was identified.

All reagents and supplies mentioned in the material section above were stored according to supplier recommendations, and their integrity was visually verified before utilization. Moreover, all reagents and supplies had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system.

The cleanroom suite used for processing procedures (CR114) comprises three interconnected sections: the innermost ISO 7 cleanroom (114B), which connects to the middle ISO 7 buffer room (114A), and then to the outermost ISO 8 anteroom (114). A positive air pressure system is maintained throughout the suite to ensure controlled, unidirectional airflow from 114B through 114A and into 114.

The innermost ISO 7 cleanroom 114B and its ISO 5 BSC E001316 were thoroughly cleaned and prepared before initiating testing by analyst Gabrielle Surber as per MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility) and MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). For Celsis aliquoting, intermediate ISO 7 buffer room 114A and its ISO 5 BSC E001798 were thoroughly cleaned and prepared by analyst Cuong Du before initiating testing as per MICRO-SOP-44 and MICRO-SOP-9. Both BSCs, E001316 in 114B and E001798 in 114A, were certified and approved by the Engineering and Quality Assurance teams prior to use.

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Phase I Summary (continued):

On 01 Sep 2026, the sample vials for ETX-260828-0527 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time. Secondary disinfection was performed in the ISO 8 anteroom (114), where the vials were again treated with acidified bleach with a 10-minute contact time before transfer into cleanroom 114B. Inside 114B, processing analyst Gabrielle Surber performed a final disinfection step with a 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001316, placed on the disinfected work surface, aseptically opened, and tested in accordance with MICRO-SOP-44. Direct inoculation was performed by adding 25 mL of sample per media container, using five 300 mL FTM jars and five 300 mL TSB jars. In accordance with suitability method record 2601120497, one gram of Bovine Serum Albumin was prepared in 100 mL of PBS, and 5 mL of this solution was added to each jar. Following testing, the media bottles were transferred into incubators E001356 and E001357 to initiate incubation.

Upon completion of incubation on 08 Sep 2026, the media bottles for ETX-260828-0527 were disinfected and transferred to the middle ISO 7 buffer room (114A) for the aliquoting step per MICRO-SOP-44 (Celsis Sterility Testing). In cleanroom 114A, the media bottles were disinfected one more time before transfer into the ISO 5 BSC E001798 located in cleanroom 114A. In ISO 5 BSC E001798, the sample was aliquoted into assay cuvettes by analyst Cuong Du. Following aliquoting, Celsis Sterility Reading was performed in accordance with MICRO-SOP-44 by analyst America Alanis.

Following the reading, sample ETX-260828-0527 was found to yield a positive reading in one of the 300 mL FTM media jars (ETX-260828-0527-3/5). The average Relative Luminescence Units (RLU) from the duplicate reading tube, originating from the FTM sample bottle, yielded 7,190 RLU, which exceeded the method cutoff of 2,955 RLU, where the FTM negative control was 985 RLU. All other FTM and TSB sample bottles in the same batch tested negative. The %CV from the duplicate reading tubes for the positive FTM bottles was well within the specification (< 30%). Additionally, all Daily Controls, including the Instrument Blank, Reagent Blank, and ATP Positive Control, were within the defined specifications, each with a %CV below 30%.

Following the OOS result, the positive FTM bottle for ETX-260828-0527 was submitted for Differential Staining and Microbial Identification under ETX-260908-0593. No microbial growth was recovered by subculture, with zero CFU observed on the plate as shown in Table 1, and no cellular morphology was visualized by differential staining. Therefore, no organism was recovered for identification or further microbiological comparison. The lack of recovery on subculture does not invalidate the initial analytical Celsis result and may be consistent with low-level or stressed microbial cells under the test conditions, though a specific physiological state could not be confirmed.

The culture media utilized were within their expiry period. The negative culture media bottles for the direct inoculation method for the original culture were handled, processed, and incubated in a manner identical to that of actual samples. No microbial growth was observed in the corresponding negative control.

Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 Biosafety Cabinets for CR114 was performed on 30 Aug 2026, as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). It was documented that all H2O2 indicators passed. This confirms the efficient monthly cleaning of all three parts of Cleanroom 114.

Attached Tables 2 and 3 present the environmental monitoring results for the duration of testing. The environmental monitoring (EM) plates were incubated for no less than 48 hours at 30-35°C and no less than an additional five days at 20-25°C as per MICRO-SOP-2 (Environmental Monitoring of the Cleanroom Facility). Table 2 pertains to Environmental Monitoring performed during Celsis Sterility Processing, and Table 3 pertains to Environmental Monitoring performed during Celsis Sterility Aliquoting.

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Phase I Summary (continued):

A review of the Environmental Monitoring (EM) results for the relevant Celsis Sterility processing period showed no microbial growth on personnel monitoring plates, settling plates, or ISO 5 BSC E001316 surface sampling plates on the date of testing (01Sep2026), the preceding sampling date (31Aug2026), or the subsequent sampling date (02Sep2026). Weekly active air and surface sampling of the cleanroom suite associated with processing was performed on 04Sep2026. No microbial recovery was observed from weekly surface sampling of the cleanroom suite. Active air sampling showed recovery of 1 CFU (ETX-260914-0487) in the ISO 8 area, identified as Gram-positive coccobacilli and Gram-positive rods, with zero recovery observed in the ISO 7 buffer or cleanrooms.

After reviewing the Environmental Monitoring (EM) results for the relevant Celsis Sterility aliquoting period, no microbial growth was observed on personnel monitoring plates, ISO 5 BSC E001798 surface sampling plates, or settling plates on the date of aliquoting (08Sep2026), the preceding sampling date (07Sep2026), or the subsequent sampling date (09Sep2026). Weekly active air sampling of the cleanroom suite performed on 10Sep2026 showed recovery of 1 CFU (ETX-260921-0520) in the ISO 8 area, identified as Gram-positive cocci, with zero recovery in the ISO 7 buffer or cleanrooms, and weekly surface sampling of the cleanroom suite showed no growth.

Evaluation of Environmental Monitoring Results: It is important to note that, for both processing and aliquoting activities, the samples are processed within validated ISO 5 BSCs (E001316 and E001798 respectively). Although microbial recovery was observed from weekly active air monitoring in the outermost ISO 8 room 114 background environment in cleanroom suite (CR114), these recoveries were not obtained from the ISO 5 processing environment or the ISO 7 cleanroom environment.

The absence of microbial recovery from both the processing and the aliquoting analyst glove plates, ISO 5 BSC surface monitoring plates, and ISO 5 settling plates indicates that no identifiable contamination transfer pathway existed between the monitored background cleanroom environment and the ISO 5 critical processing areas. Samples were also prepared and transferred into disinfected, lidded containers and transported on carts within the controlled cleanroom environment. Therefore, while the active air results demonstrate microbial recovery in the surrounding ISO 8 area, there is no evidence that the recovery originated from, or affected, the ISO 5 BSC environment where the test sample was handled and processed.

All analysts confirmed full compliance with cleaning procedures as outlined in MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology) and MICRO-SOP-44 (Celsis Sterility Testing). A review of the available data confirms that the cleanroom and equipment conditions remained within acceptable parameters. No deviations or obvious signs of laboratory error during processing and aliquoting were noted from analysts.

No other sample processed by analyst Gabrielle Surber on 01 Sep 2026 and aliquoted by analyst Cuong Du on 08 Sep 2026, failed Celsis Sterility testing that day. To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed in the same batch. All other samples processed by the same analyst on the day of testing were found to test negative for microbial growth. These findings suggest that cross-contamination between samples is unlikely.

Analyzing a 6-month sample history for Optimal Balance Pharmacy indicates, this specific analyte "MOTs-C 10 MG/ML (5 ML) Injection" has had no prior failures using the Celsis Sterility testing during this period.

Based on the laboratory investigation, no assignable laboratory cause was identified involving the analyst, instrument, reagents, supplies, or monitored laboratory environment. The initial Celsis OOS result therefore remains valid in accordance with the applicable laboratory OOS procedure.

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Has root cause been identified as a possible laboratory error?
Note: Root cause must be conclusive, verifiable, and well-documented.

☐ Yes

☐ If retesting is ***required***, complete Phase II of the OOS (initiate Form 3.100.019.F02).

☑ No, the OOS result is valid.

Reviewer comment(s):

Prepared by (*Print*)**:** Qiyue Chen	Signature/Date: SIGN
Lab Manager/Designee (*Print*)**:** Robin Seymour	Signature/Date: SIGN

Section E – Quality Disposition

Can investigation be closed?

☐ Yes

☐ Original result is valid.

☐ No, initiate Form 3.100.019.F02.

Approved by QA Representative (*Print*)**:**	Signature/Date: SIGN

Table 1: Information for ETX-260828-0527 under investigation

Processing Analyst	Aliquoting Analyst	Sample ID	Related Microbial ID	Media with microbial growth	Microbial ID
Gabrielle Surber	Cuong Du	ETX-260828-0527	ETX-260908-0593	1 x 300mL FTM	No growth was obtained

Table 2: Environmental Monitoring from Processing Performed on 01Sep26

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMMYY)	Analyst (Initials)	Day /Week(s)	Observation*	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing								
Personnel (Left Touch and Right Touch)	Daily	31Aug26	GS	Date Before Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	01Sep26	GS	Date of Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	02Sep26	GS	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC)								
Surface Sampling of ISO 5 (4 locations)	Daily	31Aug26	GS	Date Before Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	01Sep26	GS	Date of Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	02Sep26	GS	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	31Aug26	GS	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	01Sep26	GS	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	02Sep26	GS	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling Bracketing								
Active Air Sampling of Cleanrooms	Weekly	04Sep26	ISS	Week on Testing Date	1 CFU (ISO 8 114)	ETX-260914-0487	Gram (+) coccobacilli & Gram (+) rods	None
Surface Sampling of Anteroom and Cleanroom Bracketing								
Surface Sampling of Cleanrooms	Weekly	04Sep26	ISS	Week on Testing Date	No growth	N/A	N/A	None

Table 3: Environmental Monitoring from Aliquoting Performed on 08Sep26

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMMYY)	Analyst (Initials)	Day /Week(s)	Observation*	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing								
Personnel (Left Touch and Right Touch)	Daily	03Sep26	CCD	Date Before Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	08Sep26	CCD	Date of Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	15Sep26	CCD	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC)								
Surface Sampling of ISO 5 (4 locations)	Daily	07Sep26	ALA	Date Before Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	08Sep26	CCD	Date of Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	09Sep26	ALA	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	07Sep26	ALA	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	08Sep26	CCD	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	09Sep26	ALA	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling Bracketing								
Active Air Sampling of Cleanrooms	Weekly	10Sep26	SMO	Week on Testing Date	1 CFU (ISO 8 114)	ETX-260921-0520	Gram (+) cocci	None
Surface Sampling of Anteroom and Cleanroom Bracketing								
Surface Sampling of Cleanrooms	Weekly	10Sep26	SMO	Week on Testing Date	No growth	N/A	N/A	None